

BUSINESS ANALYTICS PROJECT REPORT

Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch

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Register Number	TU-6253210211004
Week	Week 1
Subject	Business Analytics
Task Name	Mathematical Customer Lifetime Value (CLV) Projection via BG/NBD Model from Scratch

Task Description

Students must implement the mathematical likelihood function of the BG/NBD model from first principles. They are required to write vectorized numerical optimization routines using SciPy's L-BFGS-B algorithm to estimate transaction rate heterogeneities and dropout probability shapes over millions of purchase transaction logs without utilizing pre-built marketing packages.

1. Project Objective

The objective is to implement a mathematical Customer Lifetime Value projection pipeline using the Beta-Geometric/Negative Binomial Distribution (BG/NBD) model from first principles. The implementation estimates heterogeneous transaction rates and dropout probabilities directly from customer transaction-history summaries.

The project emphasizes vectorized numerical computation, stable likelihood evaluation, constrained optimization, and scalability to very large transaction logs without relying on pre-built marketing-model packages.

2. Customer-Level Input Representation

The BG/NBD likelihood can be evaluated from customer-level summary statistics derived from transaction logs. A practical summary contains the customer's frequency, recency, and observation period.

For customer i :

x_i = number of repeat transactions
 t_{x_i} = time of the customer's last transaction
 T_i = total observation period

Typical model inputs:

frequency (x_i)
recency (t_{x_i})
customer age / observation time (T_i)

The original task does not specify a particular transaction dataset or time unit. The implementation should document the chosen observation window and time units.

3. BG/NBD Model Parameters

The model represents transaction-rate heterogeneity with Gamma-distributed purchase rates and dropout behavior with a Beta-distributed probability. The parameters are estimated from observed customer transaction histories.

Transaction-rate heterogeneity:

$$\lambda_i \sim \text{Gamma}(r, \alpha)$$

Dropout probability:

$$p_i \sim \text{Beta}(a, b)$$

Parameters to estimate:

r, α, a, b

Constraints:

$$r > 0$$

$$\alpha > 0$$

$$a > 0$$

$$b > 0$$

These four positive parameters define the population-level heterogeneity in purchasing intensity and customer dropout behavior.

5. Log-Likelihood and Numerical Stability

Direct multiplication of likelihood terms can underflow when millions of customer records are processed. A log-likelihood formulation is therefore preferred.

$$\begin{aligned}\log L_{\text{total}}(\theta) \\ &= \sum_i \log L_i(\theta)\end{aligned}$$

where:

$$\theta = [r, \alpha, a, b]$$

Use stable special functions such as:

$$\begin{aligned}\text{gamma}(\ln(z)) &= \log \Gamma(z) \\ \text{beta}(\ln(a,b)) &= \log B(a,b)\end{aligned}$$

Objective for optimization:

$$\begin{aligned}\text{negative_log_likelihood}(\theta) \\ &= - \sum_i \log L_i(\theta)\end{aligned}$$

Stable log-domain calculations reduce numerical overflow and underflow and make the optimization process more reliable.

6. Vectorized Optimization with L-BFGS-B

The likelihood computation should operate on arrays of customer summaries rather than processing customers through slow Python-level loops. SciPy's L-BFGS-B algorithm can then optimize the four positive parameters under explicit bounds.

```
from scipy.optimize import minimize

def objective(params, x, tx, T):
    r, alpha, a, b = params
    # vectorized log-likelihood calculation
    # return negative total log-likelihood
    return nll

result = minimize(
    objective,
    x0=initial_params,
    args=(frequency, recency, T),
    method="L-BFGS-B",
    bounds=[
        (1e-8, None),
        (1e-8, None),
        (1e-8, None),
        (1e-8, None)
    ]
)
```

L-BFGS-B is suitable for bounded numerical optimization and can enforce positive parameter bounds while using gradient-based updates.

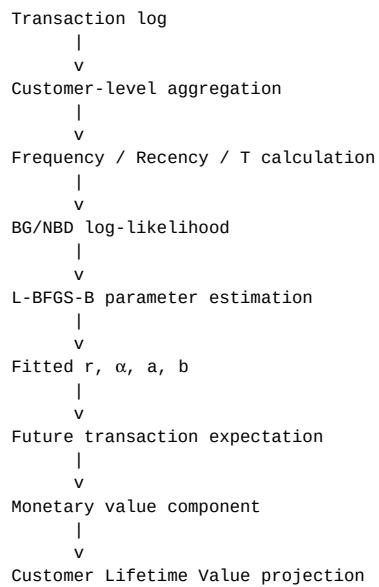
7. Scaling to Millions of Transaction Logs

The raw transaction logs should first be transformed into compact customer-level summaries. This reduces the repeated likelihood workload and makes vectorized numerical operations practical for large datasets.

Stage	Scalable Approach
Raw transactions	Read and aggregate transaction records
Customer summaries	Compute frequency, recency, and observation time
Likelihood	Vectorized NumPy operations
Special functions	Stable log-Gamma and log-Beta functions
Optimization	SciPy L-BFGS-B
Validation	Check convergence and parameter bounds
Projection	Estimate future transaction behavior / CLV inputs

8. CLV Projection Pipeline

Once BG/NBD parameters are estimated, the fitted model can support future purchase-frequency estimation. A monetary-value model can be combined with the expected future transactions when the project requires a monetary CLV estimate.



The task specifically focuses on the mathematical BG/NBD likelihood and parameter estimation. Any monetary-value component should be added as a separate model if required.

9. Validation and Performance Evaluation

Validation should include optimizer convergence, positive parameter constraints, likelihood improvement from initial parameters, and comparison of predicted behavior against held-out or known transaction patterns when available.

Performance evaluation should record processing time, memory usage, number of customers, optimization iterations, and objective-function evaluations. Vectorization should be compared with a small-loop implementation on a test subset to verify numerical consistency.

Important: The supplied task does not provide a dataset, numerical parameter values, or expected model metrics, so this report does not invent final fitted results.

10. Expected Deliverables and Conclusion

Expected Deliverables:

1. Transaction-log preprocessing and customer summary generation.
2. First-principles BG/NBD likelihood implementation.
3. Log-space numerical stability functions.
4. Vectorized objective function.
5. L-BFGS-B parameter estimation for r , α , a , and b .
6. Scalability and runtime analysis for large datasets.
7. Future transaction / CLV projection component where required.
8. Validation results and project report.

Conclusion

This Week-1 Business Analytics task implements the mathematical foundation of Customer Lifetime Value projection using the BG/NBD model. The project focuses on deriving the likelihood from first principles, performing stable vectorized calculations, estimating transaction-rate and dropout heterogeneity with L-BFGS-B, and building a scalable foundation for future customer-value projections.